

Tenure segmentation, family space, and fertility

Two local results from the existing preference specification

Claim. A rental-size cap can lower the value of parenthood and, near the point where parents first reach it, amplify the fertility cost of uncertainty. Both results hold with freely adjustable rental housing and zero transaction costs. Ownership finance determines when this restricted rental option matters in the full model.

1. A conditional rental problem

Let $m \in \{0, 1\}$ denote childlessness or one child at home. At a future date, allocate expenditure x between consumption c and physical space h , at rent $r > 0$:

$$J_m^H(x) = \max_{\substack{c+rh=x, c>0 \\ mh_p < h \leq H}} \frac{e_m^{\sigma-1} [c^\alpha (h - mh_p)^{1-\alpha}]^{1-\sigma}}{1-\sigma}. \quad (1)$$

Assume $0 < \alpha < 1$, $\sigma > 1$, $e_0 = 1$, $e_1 \geq 1$, $H > h_p > 0$, and $x > rh_p$. Here H is the rental cap and h_p the parenthood floor. This is a building block of the dynamic problem: expenditure and the rental branch are held fixed for the comparison. Define the birth advantage $D_H(x) = B + J_1^H(x) - J_0^H(x)$, where B collects fixed child benefits and costs.

Without the rental cap, the first-order condition gives

$$h_m^*(x) = \frac{(1-\alpha)x}{r} + \alpha mh_p, \quad x_1 = \frac{r(H - \alpha h_p)}{1-\alpha} < x_0 = \frac{rH}{1-\alpha}. \quad (2)$$

Thus $x_1 < x < x_0$ identifies a concrete region in which desired parental space exceeds the rental menu while desired childless space remains available.

Proposition 1. Rental expansion raises the conditional birth advantage

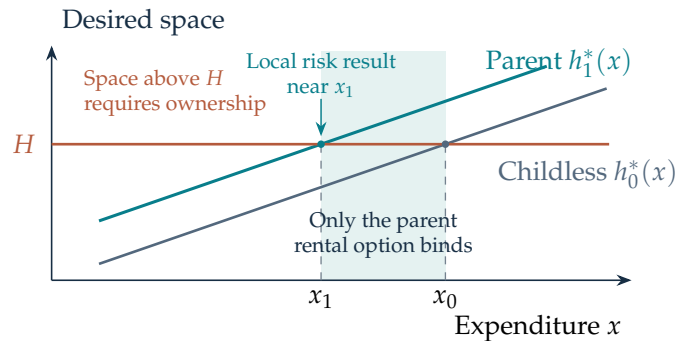
Let ζ_m be the multiplier on $h \leq H$, zero when the cap is slack. Then

$$\frac{\partial D_H(x)}{\partial H} = \zeta_1 - \zeta_0 \geq 0, \quad \text{strictly if } x > x_1. \quad (3)$$

Proof. Parents reach the cap first. If both branches bind, they share $c = x - rH$, and

$$\zeta_m = u_{c,m} \left[\frac{(1-\alpha)c}{\alpha(H - mh_p)} - r \right].$$

Parenthood raises both the bracket and marginal utility $u_{c,m}$ under (1), so $\zeta_1 > \zeta_0$. The remaining cases follow from (2). This signs a space margin from primitives, without assuming the desired sign of the birth-value response.



Reading the diagram. The shaded interval isolates the parent-only rental restriction. Sizes above H require an owner option. At the supplied benchmark, $H = 6$, $h_p = 2.585$, $\alpha = .733$, and $r = .1405$ imply $x_1 \simeq 2.160$ and $x_0 \simeq 3.157$ per four-year period. These are expenditure thresholds, not income cutoffs.

2. A local precautionary effect from the missing rental margin

Suppose the child commitment precedes future expenditure X . Let D_∞ denote the rental problem without a cap, and define its advantage over the segmented menu as $L(x) = D_\infty(x) - D_H(x)$. Near x_1 , the childless branch is slack, so $L = J_1^\infty - J_1^H$. Put $t_1 = x_1 - rh_p$ and $\lambda = J_{1,x}^\infty(x_1) > 0$. Values and slopes agree at the onset: $L(x_1) = L'(x_1) = 0$. Since capped consumption there is αt_1 ,

$$J_{1,xx}^\infty(x_1) = -\frac{\sigma\lambda}{t_1}, \quad J_{1,xx}^H(x_1^+) = -\frac{[1 + \alpha(\sigma - 1)]\lambda}{\alpha t_1}, \quad L''(x_1^+) = \frac{1 - \alpha}{\alpha} \frac{\lambda}{t_1} > 0. \quad (4)$$

Proposition 2. Segmentation locally raises the cost of risk

The loss L is convex in a sufficiently small neighborhood of x_1 , with $L = 0$ just below it. For a mean-preserving risk supported in that neighborhood, the cap adds to the risk penalty in the birth advantage by

$$\underbrace{[D_H(\bar{x}) - \mathbb{E}D_H(X)] - [D_\infty(\bar{x}) - \mathbb{E}D_\infty(X)]}_{\text{additional risk penalty from segmentation}} = \mathbb{E}L(X) - L(\bar{x}) \geq 0, \quad \bar{x} = \mathbb{E}X. \quad (5)$$

For equiprobable $X = x_1 \pm \varepsilon$, the additional penalty is $L''(x_1^+)\varepsilon^2/4 + o(\varepsilon^2) > 0$. The certainty birth advantages coincide at x_1 , so an increasing birth-choice rule gives a larger decline with the cap. Away from a common certainty baseline, (5) orders value penalties rather than changes in logit probabilities.

Economic force. Parents lose the ability to expand space in a favorable realization while childless households retain it. No moving cost or irreversible ownership choice is needed. The cap need not bind in the poorest states, and the curvature ordering can reverse farther from x_1 .

3. Ownership finance and the intergenerational connection

In the full model, sizes above six rooms exist only in the owner menu. For a nonowner considering purchase, the standard simultaneous budget implies

$$b' = M - Ph - c - k(h) \geq -\varphi Ph \implies c + k(h) + (1 - \varphi)Ph \leq M. \quad (6)$$

Here P is the price per room, φ the LTV limit, $k(h)$ other current costs, and M resources including current income and liquid assets with their return. Finance can prevent ownership from replacing missing rental space. The down payment uses jointly available resources, not only initial wealth. Existing owners can also bring housing equity to a purchase.

The rental theorem applies when both branches rent locally or the owner alternative is inactive. With endogenous tenure, holding continuation fixed, the local cap derivative is instead

$$\frac{\partial G}{\partial H} = p_R^1 \zeta_1 - p_R^0 \zeta_0, \quad (7)$$

where G is the birth advantage and p_R^m the renter-choice probability. A slack childless rental option and a binding parental option with positive renter probability suffice for a positive response. Permanent policy adds discounted future cap multipliers on both paths; saving and owner switching preclude a global sign.

What to measure. Locate states with a binding parental rental option and a slack childless option. Record the owner alternative, its financing constraint, and birth-choice sensitivity. Compare rental expansion across risk levels at common initial states and expected level earnings, including zero selling costs. Since $6 > 2.585$, the cap does not mechanically prevent parenthood. Age and wealth differences can govern access to extra space; fixed entrant wealth does not establish parent-to-own-child inheritance effects.

Sources. Inputs: supplied benchmark memo, 6 October 2026. Related mechanisms: [Sommer \(2016\)](#), *JME* 83, 27–38; [Fisher and Gervais \(2011\)](#), *IER* 52, 883–912. Equations (2)–(7) are conditional derivations here.